

4 Organisms are classified in three domains. The domain Eukarya is divided into four kingdoms.

(a) The four eukaryotic kingdoms are listed in Table 4.1.

Complete Table 4.1 by writing 'yes' or 'no' to produce a summary of some of the main features of each kingdom.

Table 4.1

kingdom	species may be unicellular	species may have cell walls	species may show autotrophic nutrition
Animalia	no		
Fungi			no
Plantae		yes	
Protoctista			

[4]

(b) Meiosis occurs in the kingdom Plantae.

Fig. 4.1 shows drawings of photomicrographs of three stages of meiosis in the lily plant, *Lilium grandiflorum*. Individual chromosomes and their structure **cannot** be seen clearly.

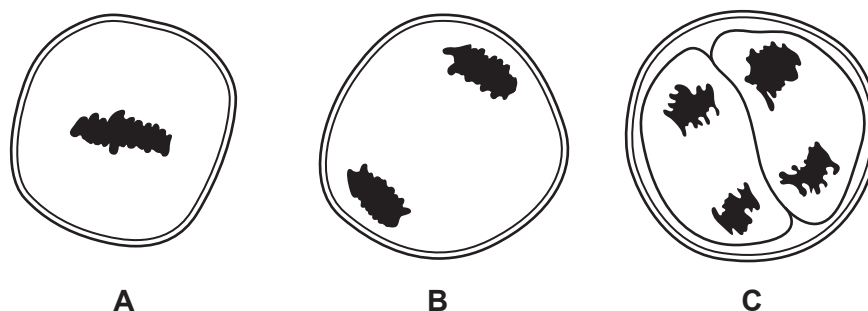


Fig. 4.1

Identify the three stages of meiosis shown in Fig. 4.1.

A

B

C

[3]

(c) The organisms in the other domains are prokaryotes.

Name the **two** domains containing prokaryotic organisms **and** describe differences between these two domains.

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..... [3]

(d) Viruses are not cellular organisms, but they are classified based on their characteristics.

Outline how viruses are classified.

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..... [2]

[Total: 12]